

CO1 - AT-2

1) Constraint - Based problem solving (Backtracking Search)

Given:-

Doctors: D_1, D_2, D_3

Shifts: Morning (M), Afternoon (A), Night (N)

Constraints

D_1 cannot work night

D_2 must work before D_3 ($M < A < N$)

only one doctor per shift

D_3 cannot work morning

Every doctor gets exactly one shift.

Backtracking steps

Step 1:-

Assign $D_1 \rightarrow$ Morning ✓

Step 2:-

Assign $D_2 \rightarrow$ Afternoon ✓

Step 3:-

Try $D_3 \rightarrow$ Morning x (not allowed)

Step 4:-

Try $D_3 \rightarrow$ Night ✓

∴ All constraints are satisfied.

Final Schedule

D_1	Morning
D_2	Afternoon
D_3	Night

2) Robot Grid Problem (BFS search)

constraints:-

Move:- Up, Down, Left, Right

Cost per move = 1

Obstacles cannot be crossed
Manhattan distance used as heuristic

BFS node Expansion

Start $\rightarrow$ Expand S

Queue:- [S]

Expand S:- [A, B]

Expand A:- [B, C]

Expand B:- [C, D]

Expand C:- [D, E]

Expand D:- [E, G]

$\therefore$ Goal G reached.

Optimal path:- $S \rightarrow A \rightarrow C \rightarrow E \rightarrow G$.

Cost:-

Number of moves = 4

Total cost = 4.

3. Analyze the Given Constraints

a) Effect of constraints

Robot moves only in four directions
obstacles must be avoided
limited sensing means it only knows nearby cell

Every move costs 1.

Risky Zones add on extra cost of 2

Robot updates its map while moving.

Objective is to reach the goal with

minimum total cost.

b) Solution Approach (Uniform cost Search / online search)

Start at S

observe adjacent cells

Calculate movement cost

Ignore blocked cells

Expand the node with lower cost

Update knowledge after every move

Repeat until G is reached.

c) AI Concepts Used.

Intelligent Agent

Senses the environment and selects

the best action

Rationality :-

Chooses the path with the minimum total cost while avoiding obstacles.

Environment Type :-

Partially type

Dynamic

Sequential

Discrete

Single-agent.

d) Justification.

Uniform Cost search is suitable because it always selects the lowest cost path, handles varying movement costs, avoids obstacles, and updates decision as new information becomes available, ensuring safe and efficient navigation.